

Technical Datasheet WIN-IO-4DOAOM

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

4Port DO-4-AO Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Power	Input Power – 12 - 24 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DO with Relay Output –

Channels	4
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10 ⁵
Mechanical Life	1 x 10 ⁷
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

AO Outputs –

Channels	4 AO Channel
Input Signal	Modbus RS485
Output Resolution	4 – 20 mA (10Bit)
Accuracy	± 2% Full scale

Configuration Settings: -

Communication Speed	9600 – 115200.
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave Id	Soft Setting
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3.
Function code AO	0x03 Read Holding Register.
AO Register Address	0,1,2,3,4,5,6,7.
Function code Soft Serial	0x03 Read Holding Register

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
0	DO1	00001	0X05,0X0F	RS485	1 Bit Boolean
1	DO2	00002	0X05,0X0F	RS485	1 Bit Boolean
2	DO3	00003	0X05,0X0F	RS485	1 Bit Boolean
3	DO4	00004	0X05,0X0F	RS485	1 Bit Boolean
1	AO 1 - 10 Bit	40001	0x03	RS485	16 Bit Unsigned int
2	AO 2 - 10 Bit	40002	0x03	RS485	16 Bit Unsigned int
3	AO 3 - 10 Bit	40003	0x03	Rs485	16 Bit Unsigned int
4	AO 4 - 10 Bit	40004	0x03	Rs485	16 Bit Unsigned int
5	AO1 - 12 Bit	40005	0x03	RS485	16 Bit Unsigned int
6	AO2 - 12 Bit	40006	0x03	RS485	16 Bit Unsigned int
7	AO3 - 12 Bit	40007	0x03	RS485	16 Bit Unsigned int
8	AO4 - 12 Bit	40008	0x03	RS485	16 Bit Unsigned int
10	Display Baud Rate (Default:960)	40011	0x03	RS485	16 Bit Unsigned int
11	Enter New Baud Rate	40012	0x03	RS485	16 Bit Unsigned int
12	Display Slave ID (Default:1)	40013	0x03	RS485	16 Bit Unsigned int
13	Enter New Slave ID	40014	0x03	RS485	16 Bit Unsigned int
14	Display Parity (Default: None-3, Odd-1, Even-2)	40015	0x03	RS485	16 Bit Unsigned int
15	Enter New Parity	40016	0x03	RS485	16 Bit Unsigned int
16	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16 Bit Unsigned int
17	Enter New Start/ Stop Bit	40018	0x03	RS485	16 Bit Unsigned int

Additional Features: -

All inputs and communication ports isolated

Input power reverse polarity safety

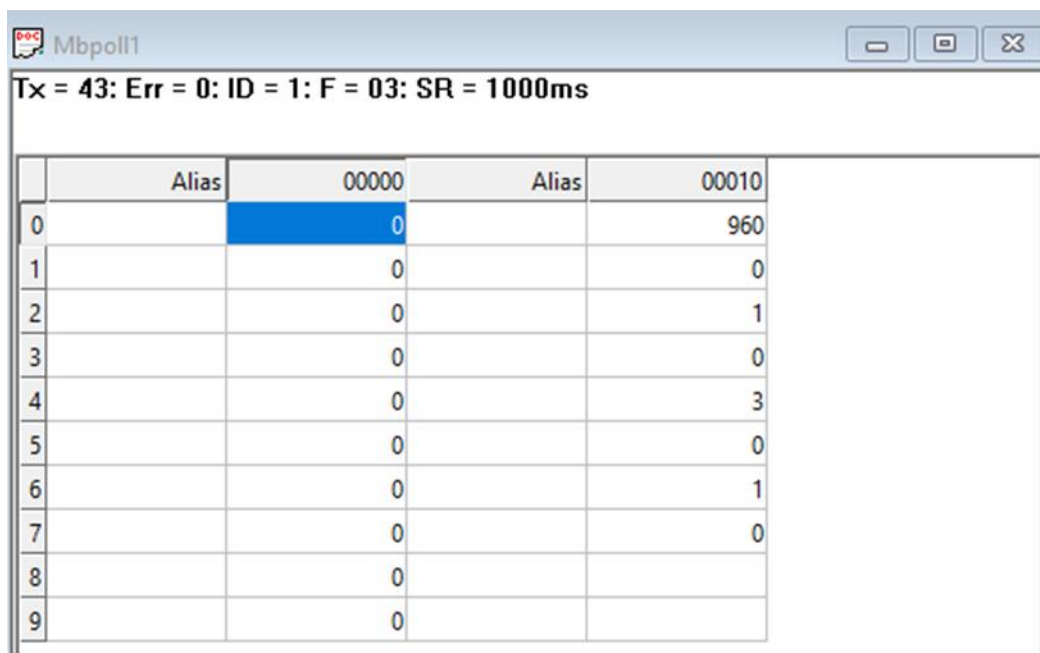
ESD Safety IEC 61000-4-2, $\pm 30\text{KV}$ contact, $\pm 30\text{KV}$ air

EFT IEC 61000-4-4, 50A (5/50ms)

750V isolation.

CRC Error check.

No configuration needed on the IO board

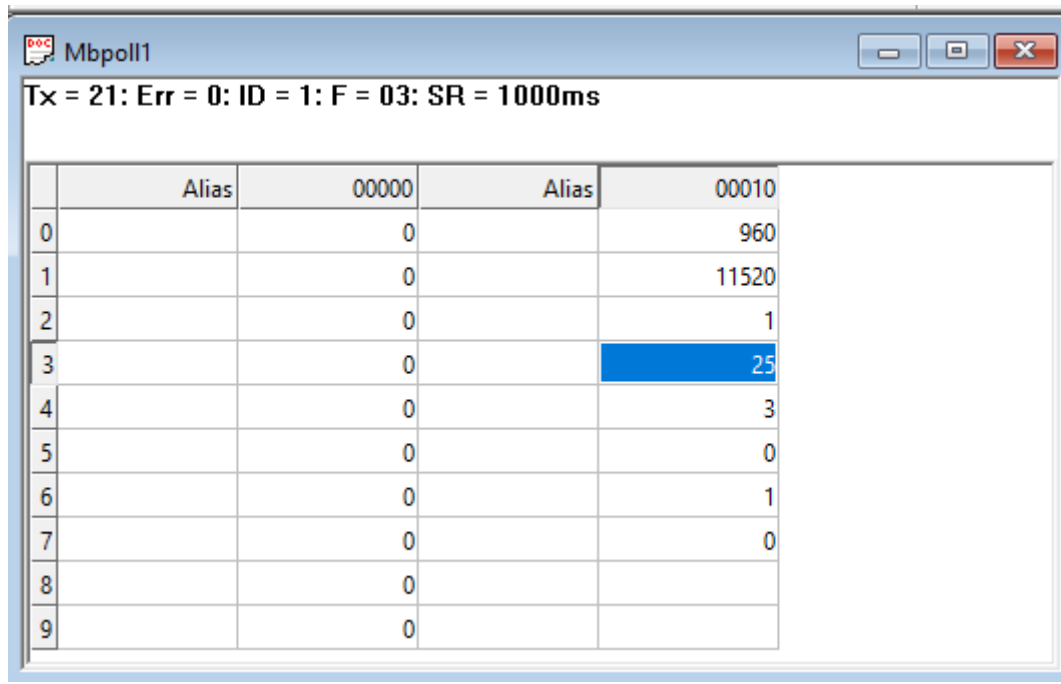


The screenshot shows a software window titled "Mbpoll1" with a status bar at the top displaying "Tx = 43: Err = 0: ID = 1: F = 03: SR = 1000ms". Below the status bar is a table with 5 columns: an index column (0-9), two "Alias" columns, and two numerical columns (00000 and 00010). The table contains the following data:

	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

Note:

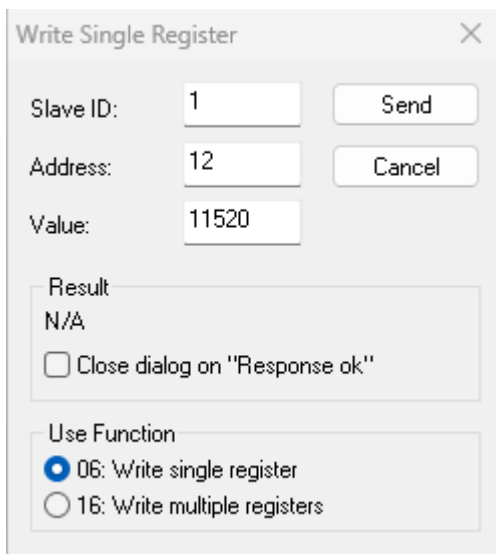
1. In register 40015- you will able to see the current parity and you can change it in register 40016
2. In register 40017- you will able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system.



	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.

To change the baud rate and slave id you have to enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below.



Write Single Register

Slave ID:

Address:

Value:

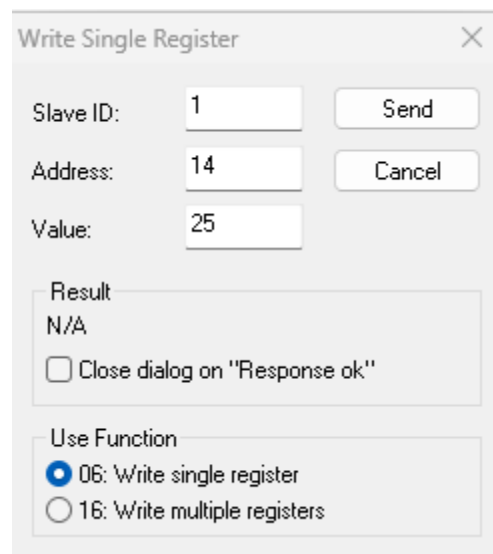
Result: N/A

☐ Close dialog on "Response ok"

Use Function:

☒ 06: Write single register

☐ 16: Write multiple registers



Write Single Register

Slave ID:

Address:

Value:

Result: N/A

☐ Close dialog on "Response ok"

Use Function:

☒ 06: Write single register

☐ 16: Write multiple registers

Mbpoll1

Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

Note:

1. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it like this 960/1920/5760/11520etc.
2. After that press the reset button or reboot the system.
3. Then you will get the present value of 11520 at 40011 baud rate(115200) and 25 at 40013 slave id.
4. You can Change Parity Bit , stop bit Of the device with same way.

Recommended Cable Electrical Characteristics: -

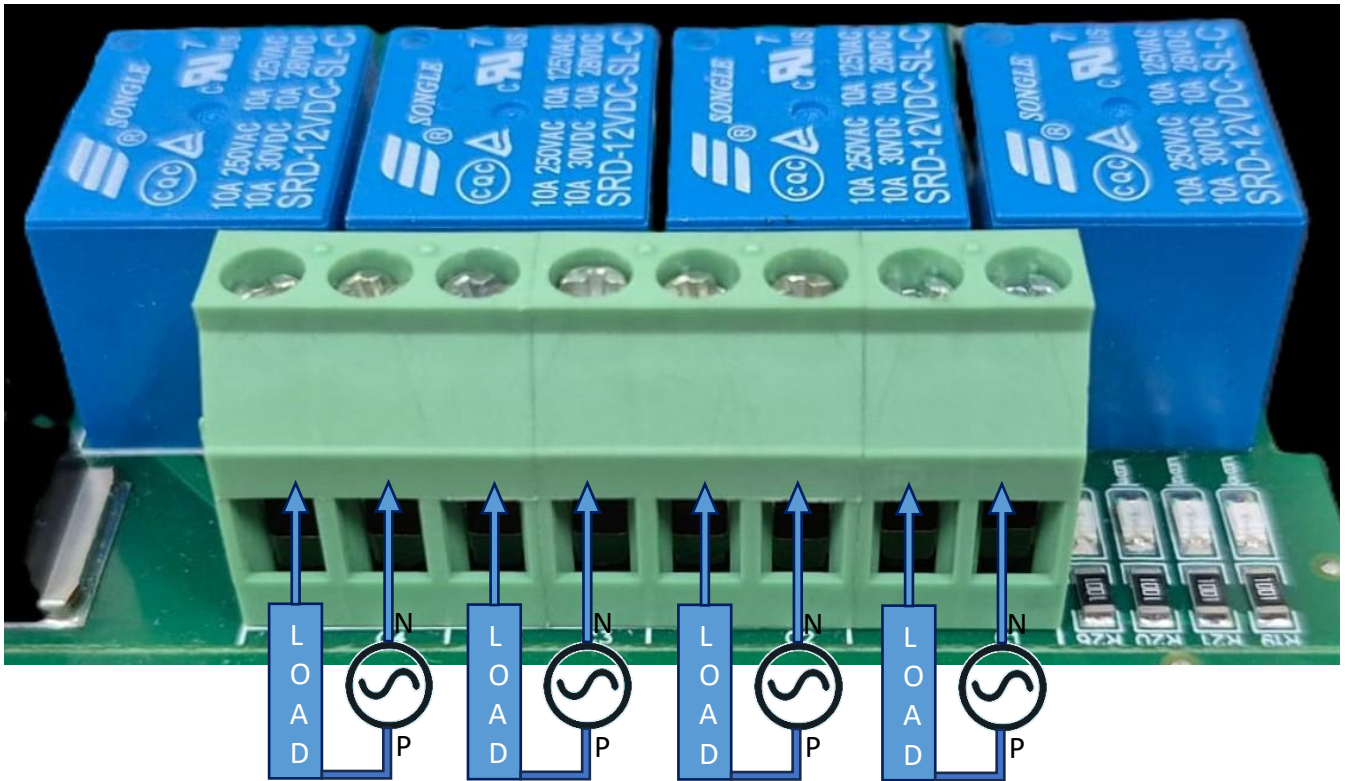
22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft



Default Setting
DIP Switch_____

NOTE – By default the Jumper setting on the board is 10 V and 10 Bit. – By default, Baud rate DIP Switch 1 is set to ON, toggling it OFF and Pressing reset button will bring the device to default RS485 Configuration i.e. Baud rate - 9600, Stop bit - 1, Parity - None, Data Bit - 8. Keep the indicated DIP switch to ON position for Setting desired configuration.

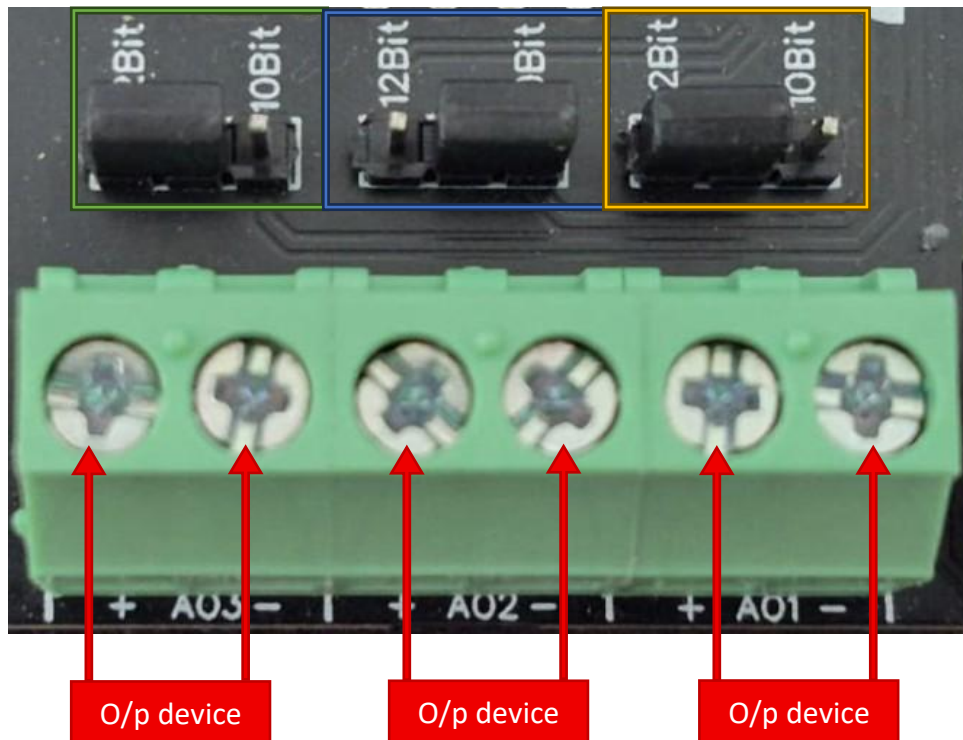
DO



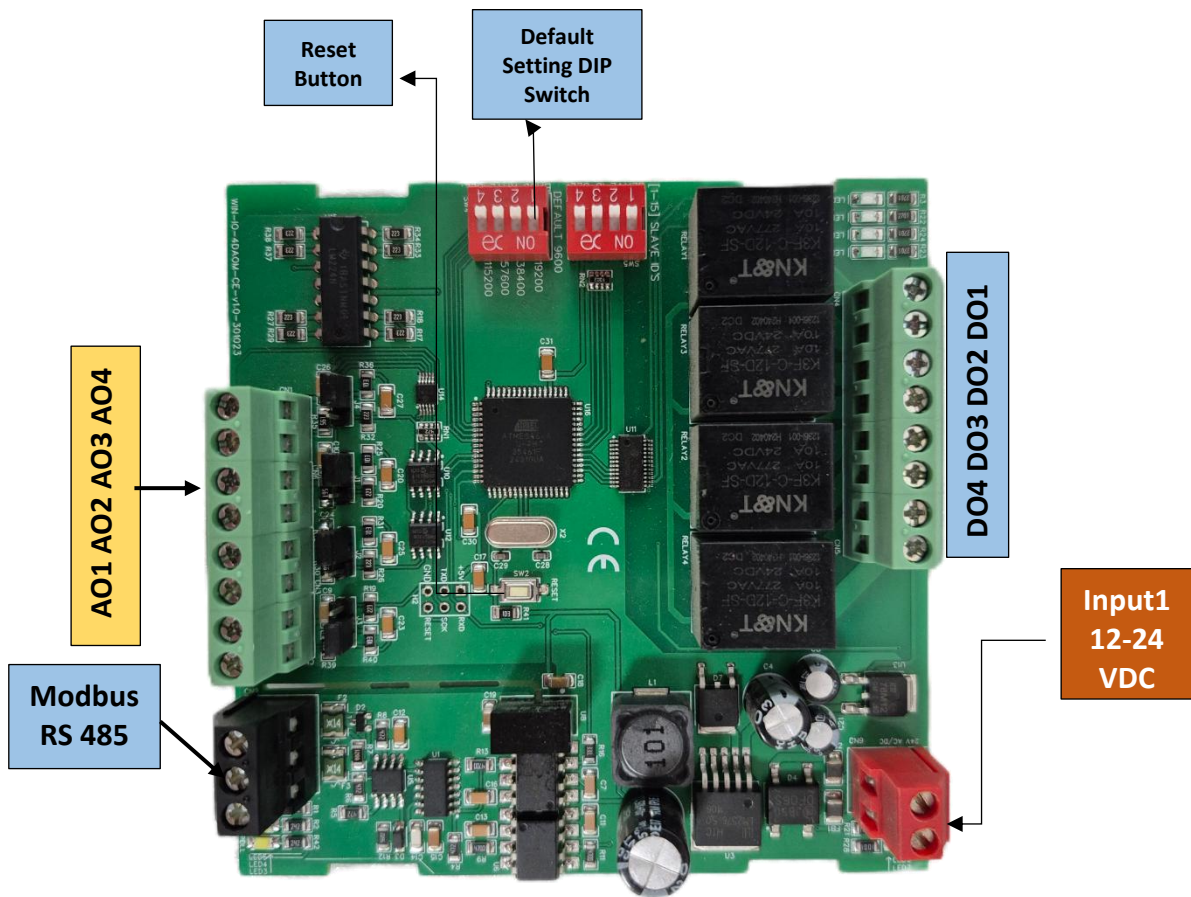
- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is **internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO** remains **open**, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

AO



- **Analog Output (AO) Operation:** Each channel provides two terminals: + (**Positive**) and - (**Negative/Common**). These are used to send variable analog signals to **external output (O/p) devices**, such as variable frequency drives or proportional valves.
- **Channel Configuration (Visual Guide):** Each of the three AO channels features a **dedicated jumper** to set the **output bit resolution**. Refer to the color-coded boxes in the board diagram to configure your desired resolution:
 - **Green & Yellow Box (12-bit):** Set the resolution jumper to **12Bit**.
 - **Blue Box (10-bit):** Set the resolution jumper to **10Bit**.
- **Output Device Wiring Connections:**
 - The external device's positive signal wire should be connected to the + terminal of the desired channel (AO1, AO2, or AO3).
 - The device's common or ground wire should be connected to the corresponding - terminal..
- **Crucial Setup Recommendation:** Always ensure that the hardware jumper setting (10-bit or 12-bit) matches the output scaling configured in your main control software. A mismatch between software resolution and hardware configuration will result in incorrect output signal levels.



Contact Information

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